MIL-I-63120A(AR)
13 February 1981
SUPERSEDING
MIL-I-63120(AR)
11 April 1977

MILITARY SPECIFICATION

INTERMEDIATE FREQUENCY/DISCRIMINATOR/SQUARE AMPLIFIER HYBRID FOR XM122 FIRING DEVICE RECEIVER

This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of microcircuit hybrid module for the XM122 firing device receiver.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assruance Provisions, General Specification for

MIL-M-55565 - Microcircuits, Packaging of

STANDARDS

MILITARY

MIL-STD-105 - Sampling, Procedures and Tables for Inspection by Attributes

MIL-STD-883 - Test Methods and Procedures for Microelectronics

FSC: 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9278752 - IF/Discriminator/Square Amplifier Hybrid

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. - The components and assembly shall comply with all requirements specified in drawing 9278752 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Metals. - External metal surfaces shall be corrosion resistant or shall be plated or treated to resist corrosion. External leads shall meet the requirements specified in 3.5.

3.4 Other Materials. - External parts, elements or coatings including markings shall be inherently non-nutrient to fungus and shall not blister, crack, outgas, soften, flow or exhibit defects that adversely affect storage, operation or environmental capabilities of hybrids delivered to this specification under the specified test conditions.

3.5 Lead Material and Finish

3.5.1 Lead Material. - Lead material shall conform to one of following chemical compositions:

Type A

Iron	53 percent, nominal
Nickel	29 ± percent
Cobalt	17 ± percent
Manganese	0.65 percent, maximum
Carbon	0.06 percent, maximum
Silicon	0.20 percent, maximum
Magnesium	0.10 percent, maximum
Zirconium	0.10 percent, maximum
Titanium	0.10 percent, maximum

(Combined total of aluminum, magnesium, zirconium, totanium to be a maximum of 0.20 percent)

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Type B

Nickel	41 - 43 percent
Manganese	0.80 percent, maximum
Silicon	0.30 percent, maximum
Carbon	0.10 percent, maximum
Chromium	0.25 percent, maximum
Cobalt	0.50 percent, maximum
Phosphorous	0.025 percent, maximum
Sulphur	0.025 percent, maximum
Aluminum	0.10 percent, maximum
Iron	Remainder

3.5.2 External Lead Finish. - Lead finish shall conform to one of the following:

a. Hot solder dip - The hot solder dip shall be homogeneous with a minimum thickness at the crest of the major flats of 200 microinches of solder (SN60 or SN63) over the primary finishes in accordance with (b) below or over nickel plate with a plating thickness of 100 microinches minimum and 200 microinches maximum.

b. Bright acid tin plate - Thickness of 100 microinches minimum and 400 microinches maximum. Optional electroless or electrolytic nickel or copper underplating, if used, shall be a minimum of 10 microinches and a maximum of 100 microinches in thickness.

NOTE: It is recognized that "bright acid tin plate", a term which refers to the process as well as the appearance, can yield a range of texture or reflectivity. It is intended that this finish be dense and continuous and that it will meet the solderability and environmental requirements of this specification.

All plating of primary finishes or underplating must be deposited in such a way that the plating is applied to a clean, non-oxidized metal surface. Where electroless undercoating is used, the device must proceed immediately to the primary finish in order to maintain solderability or the ability to overplate effectively. All finishes shall meet the solderability tests of MIL-STD-883, Method 2003.

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3.5.3 Lead Integrity. - Hybrids, when tested as specified in MIL-STD-883, Method 2004, Test Condition B1, shall exhibit no evidence of breakage, loosening, or relative motion between the lead and the hybrid body or damage to lead plating when examined at a magnification between 10X and 20X.

3.6 Design and Construction. - Hybrid design and construction shall be in accordance with the requirements specified herein and in the applicable specification or drawing.

3.6.1 Package. - All hybrids supplied under this specification shall be hermetically sealed in glass, metal or ceramic (or combinations of these) packages. No organic or polymeric materials (lacquers, varnishes, coatings, adhesives, greases, etc.) shall be used inside the hybrid package, unless approved by the design activity.

3.6.2 Marking Permanency and Legibility. - Marking shall be correct, permanent, legible and capable of withstanding the requirements of MIL-STD-883, Method 2015.

3.7 Design Documentation. - When specified in the procurement document, design, topography and schematic circuit information for all hybrids supplied under this specification shall be submitted to the procuring activity. Unless otherwise specified, all design documentation shall be sufficient to depict completely the physical and electrical construction of the hybrids supplied under this specification, and shall be traceable to the specific part, drawing or type numbers to which it applies, and to the production lot(s) and inspection lot codes under which hybrids are manufactured and tested so that revisions can be identified.

3.8 Schematic Circuits. - The actual schematic diagrams of all hybrids supplied under this specification shall be provided, showing all electrical elements functionally designed into the hybrid together with their values, when applicable. Parasitic elements, such as those associated with resistors or main isolation, which are important to the proper functioning of the hybrid shall be included but shall be depicted in such a way as to identify them as parasitic elements.

3.9 Internal Conductors. - Internal conductors (metallization stripes, contact areas, bonding interfaces, wires or other current carrying elements) of the hybrids shall be designed so that no properly fabricated conductor shall experience, in normal operation, a current density in excess of 2×10^5 amperes/ cm^2, including allowances for worst case conductor compositions and cross sectional area, normal production tolerances on design dimensions, and actual thicknesses at critical areas such as steps in elevation or contact windows.

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3.10 Screening. - All modules to be delivered in accordance with this specification shall have been subjected to, and passes, all of the following screening requirements in sequence.

3.10.1 Internal Visual (Pre-Seal). - Hybrids shall meet the requirements of MIL-STD-883, Method 2017.

3.10.2 Bond Strength. - Hybrid bond, when tested as specified in MIL-STD-883, Method 2011, Test Condition D shall exhibit a minimum bond strength of 2 grams.

3.10.3 Pre-Seal Bake. - The hybrids shall be baked at 150°C in a dry nitrogen atmosphere for a minimum of two hours immediately prior to sealing in a dry nitrogen atmosphere. As an alternate, baking may be performed in a 50 x 10^{-3} Torr vacuum. The dry nitrogen atmosphere shall contain 50 ppm or less of moisture.

3.10.4 Stabilization Bake. Capped hybrids, after being subjected to Stabilization Bake in accordance with MIL-STD-883, Method 1008, Test Condition C. for 48 hours, shall meet the electrical requirements of 3.11.

3.10.5 Temperature Cycling. - Hybrids, when tested as specified in MIL-STD-883, Method 1010, Test Condition C, shall meet the electrical requirements of 3.11.

3.10.6 Constant Acceleration. - Hybrids, when tested in accordance with MIL-STD-883, Method 2001, Test Condition A, Y1 axis shall meet the electrical requirements of 3.11.

3.10.7 Fine Leak. - Hybrids shall meet the requirements specified in MIL-STD-883, Method 1014. Test Condition A1.

3.10.8 Gross Leak. - Hybrids shall meet the requirements specified in MIL-STD-883, Method 1014. Test Condition C.

3.10.9 Burn-In. - Hybrids shall be subjected to burn-in testing in accordance with Figure 3 and MIL-STD-883, Method 1015, Test Condition B, for 160 hours at 125°C minimum, and shall meet the electrical requirements of 3.11 after exposure. When approved by the procuring activity, an optional pre-seal burn-in procedure in accordance with MIL-STD-883, Method 5008, Paragraph 3.1.3.1 may be used. When used, pre-seal burn-in shall immediately precede the pre-seal bake operation of 3.10.3.

3.10.10 External Visual. - Hybrids shall meet the requirements of MIL-STD-883, Method 2009. There shall be no evidence of: nonconformance with the detailed drawings, the absence of any required feature, damage, corrosion or contamination.

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3.11 Electrical Performance Characteristics

3.11.1 Intermediate Frequency (IF) Voltage Gain.- The intermediate frequency (IF) voltage gain measured between the test fixture input J1 and the IF output (J5 of the test fixture) shall be minus six and five tenths plus or minus three and five decibels (-6.5 ± 3.5 db) when tested in accordance with 4.5.1.1.

3.11.2 Discriminator Filter Characteristics. - The discriminator filter response at pin 3 shall be minus three decibels (-3db) within the frequency range of five hundred fifty and eight hundred fifty Hertz (550 and 850 Hz) with respect to the response at twenty five Hertz (25Hz) when it is tested in accordance with 4.5.1.2.

3.11.3 Squaring Amplifier Gain.- The squaring amplifier gain shall be thirty two decibels plus or minus three decibels (32 db ± 3 db) when tested in accordance with 4.5.1.3.

3.11.4 Squaring Amplifier Characteristics. - The module shall produce a signal at the squaring amplifier output, pin two (2), with the following characteristics when tested in accordance with 4.5.1.4:

a. The output signal shall be a square wave with a minimum peak to peak amplitude of one hundred millivolts(100 mv).

b. The symmetry of the output signal shall be between forty percent and sixty percent (40% and 60%) of a full cycle.

c. The symmetry of the output signal shall be between forty percent and sixty percent (40% and 60%) of the full cycle when the signal frequency is reduced to fifty hertz (50 Hz).

d. There shall be an inversion of data from pin three (3) to pin two (2).

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3.11.5 Current Drains. - The maximum current drains shall be forty microamperes and one and five tenths milliamperes (40 μa and 1.5 ma), respectively, from the one and one hundred and fifty five thousandths volt and the three and twenty five hundredths volt (1.155 V and 3.25 V) supplies when tested as specified in 4.5.1.5.

3.11.6 Input Capacitances. - The module input capacitance, measured between pin one (1) and pin sixteen (16) and between pin thirteen (13) and pin fifteen (15), shall be sixty plus or minus three picofarads (60 ± 3 pfd) and sixty seven plus or minus three picofarads (67 ± 3pfd) respectively, when tested in accordance with 4.5.16.

3.11.7 Voltage Variations. - The module shall meet the requirements 3.11.1 through 3.11.5 when all referenced voltages are varied to specified maximum and minimum limits as tested in 4.5.1.7.

3.12 Temperature.

3.12.1 Test Temperature. - Unless specified otherwise, all testing shall be performed at a temperature of twenty five degrees plus or minus five degrees centigrade (25°C ± 5°C).

3.12.2 Temperature Variations. - The module shall meet the requirements of 3.11.7 over the temperature range of minus thirty two degrees centigrade (-32°C) to plus sixty five degrees centigrade (+65°C) when tested as specified in 4.5.2.

3.13 Vibration. - Hybrids, when tested as specified in MIL-STD-883, Method 2007, Test Condition A, shall meet the electrical requirements of 3.11 after exposure. Each sample unit will be opened and internal visual examination performed at 30X magnification. No evidence of damage to package, leads, seals and markings is permitted.

3.14 Data Requirements. - Data shall be recorded and retained for all tests required by this specification and as specified in 4.5.4.

3.15 Rework.

3.15.1 Rework Provisions. - Rework of the package seal for hybrid microcircuits shall be permitted a maximum of two times after final seal and only for packages which have an outer seal perimeter of at least two inches in total length. Allowable rework includes recleaning of any hybrid microcircuit or portion thereof, rebranding to correct defective marking and lead straightening (provided the reworked devices meet the requirements of Paragraph 3.10.10 for conditions of leads), unless otherwise specified. Any module failing to comply with all of the test requirements and undergoes rework or repair shall be tested as specified in 4.5.3.

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3.15.2 Rebonding and Element Replacement for Hybrid Microcircuits. - Unless otherwise specified, rebonding and replacement of attached elements in hybrid microcircuits shall be permitted with the following limitations:

a. No scratched, open or discontinuous metallization paths or conductor patterns shall be repaired by bridging with or addition of bonding wire or ribbon.

b. No more than one rebond attempt shall be permitted at any pad or post and no rebonds shall be made directly over an area where metallization of intended wire bond areas has been lifted.

c. The total number of rebond attempts (exclusive of total element replacement) shall be limited to a maximum of 10 percent of the total number of bonds in the hybrid microcircuits. The 10 percent limit on rebonds may be interpreted as the nearest whole number to the 10 percent value. A bond shall be defined as a wire to post or wire to pad bond (i.e., for a 14 lead wire-bonded package there are 28 bonds). Bond-offs required to clear the bonder after an unsuccessful first bond attempt need not be considered as rebonds provided they can be identified as bond-offs by being made physically off the plated post or if they contain a non-typical number of wedge marks. The initial bond attempt need not be visible. The replacement of one wire bonded at one end or an unsuccessful bond attempt at one end of the wire counts as one rebond; a replacement of a wire bonded at both ends, or an unsuccessful bond attempt of a wire bonded at both ends, or an unsuccessful bond attempt of a wire already bonded at the other end counts as two rebonds. A ball bond on top of a ball is not permissible.

d. Replacement of active or passive elements or dice shall be limited to a maximum of three rework cycles at any given process station with a limit of two replacements of each element or die, except that there shall be no more than 10% replacement of the total elements and/or dies. Rebonding of wires to substrate pad or package post done to effect element replacement shall not be counted against the rebond limitations of 3.15.2c.

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e. Regardless of the number of allowable rebonds or element replacements, all rebonding or element replacement of the hybrid must be accomplished in no more than three recycles of any single hybrid for rework after initial electrical test.

3.16 Workmanship. - The module shall be free from dirt, chips, grease, flux, rust or other foreign matter that may impair proper functioning. The cleaning methods used shall not be injurious to any part or assembly nor shall the parts or assembly be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of Inspection. - The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall be selected from parts that have passed the tests of 4.4.2.1. Each first article sample shall consist of the following items in the quantities as indicated:

Part Number	Description	Quantity
9278752	Hybrid, I.F./Discriminator/Squaring Amplifier	25
9278752	Hybrid, I.F./Discriminator/Squaring Amplifier (Unsealed)	10

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE I FIRST ARTICLE INSPECTION

PARAGRAPH	TITLE			SHEET 1 OF 2	DRAWING NUMBER
	HYBRID, I.F./DISCRIMINATOR/SQUARE AMPLIFIER				9270752
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	GROUP I (a)	25			
	Voltage Variations			3.11.7	4.5.1.7
	Temperature Varations			3.12.2	4.5.2
	GROUP II (a)	5			
	Physical Dimensions			3.2	Gage
	Marking Permanence			3.6.2	MIL-STD-883
	External Visual	3		3.10.10	MIL-STD-883
	Solderability	(b)		3.5.2	MIL-STD-883
	Lead Integrity			3.5.3	MIL-STD-883
	Fine Leak			3.10.7	MIL-STD-883
	Gross Leak			3.10.8	MIL-STD-883

NOTES: (a) Each unit in the group shall be subjected to all the examinations or tests in sequence for that group unless otherwise specified. (b) 10% of the leads on each unit shall be tested.

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TABLE I FIRST ARTICLE INSPECTION (CON'T)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	HYBRID, I.F./DISCRIMINATOR/SQUARE AMPLIFIER				9278752
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	GROUP III (a)	10			
	Vibration			3.13	MIL-STD-883
	Constant Acceleration			3.10.6	MIL-STD-883
	Fine Leak			3.10.7	MIL-STD-883
	Gross Leak			3.10.8	MIL-STD-883
	I.F. Voltage Gain			3.11.1	4.5.1.1
	Discriminator Filter Characteristics			3.11.2	4.5.1.2
	Squaring Amplifier Gain			3.11.3	4.5.1.3
	Squaring Amplifier Characteristics			3.11.4	4.5.1.4
	Current Drains			3.11.5	4.5.1.5
	Input Capacitance			3.11.6	4.5.1.6
	GROUP IV (unsealed) (a)	10			
	Internal Visual			3.10.1	MIL-STD-883
	Bond Strength	5	(c)	3.10.2	MIL-STD-883

NOTES (c) The AQL (p) shall be .65 or less when the computed probability of acceptance (PA) is .95 using the total number of bond breakages observed (c) while testing total number of bonds (n) in the five sample hybrids, in accordance with the expression $PA = e^{-np} \sum_{x=0}^{c} \frac{(np)^x}{x!}$ for a Poisson distribution as an approximation to the binomial distribution.

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination (See MIL-A-48078). - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-Std-105, Inspection Level II.

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	HYBRID, I.F./DISCRIMINATOR/SQUARE AMPLIFIER				9278752
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Internal Visual		100%	3.10.1	4.4.3.1, MIL-STD-883
102	Bond Strength		(a)	3.10.2	4.4.3.1, MIL-STD-883
103	Pre-Seal Bake		100%	3.10.3	4.4.3.1, MIL-STD-883
104	Stabilization Bake		100%	3.10.4	4.4.3.1, MIL-STD-883
105	Temperature Cycling		100%	3.10.5	4.4.3.1, MIL-STD-883
106	Constant Acceleration		100%	3.10.6	4.4.3.1, MIL-STD-883
107	Fine Leak		100%	3.10.7	4.4.3.1, MIL-STD-883
108	Gross Leak		100%	3.10.8	4.4.3.1, MIL-STD-883
109	Burn-In		100%	3.10.9	4.4.3.1, MIL-STD-883
110	External Visual		100%	3.10.10	4.4.3.1, MIL-STD-883
111	I.F. Voltage Gain		100%	3.11.1	4.5.1.1
112	Discriminator Filter Characteristics		100%	3.11.2	4.5.1.2
113	Squaring Amplifier Gain		100%	3.11.3	4.5.1.3
114	Squaring Amplifier Characteristics		100%	3.11.4	4.5.1.4
115	Current Drains		100%	3.11.5	4.5.1.5
116	Input Capacitance		100%	3.11.6	4.5.1.6

NOTES: (a) See 4.4.3.1 (b) 10% of the leads on each unit shall be tested.

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PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.1	HYBRID, I.F./DISCRIMINATOR/SQUARE AMPLIFIER		9278752
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR					
117	Pin Location		0.40%	3.2	Gage
118	Pin Length		0.40%	3.2	Gage
119	Package Length		0.40%	3.2	Gage
120	Package Width		0.40%	3.2	Gage
121	Assembly Damaged or Distorted		0.40%	3.2	Gage
122	Marking Permanency	5		3.6.2	MIL-STD-883
123	Solderability (b)	10		3.5.2	MIL-STD-883
124	Lead Integrity	10		3.5.3	MIL-STD-883
MINOR					
201	Pin Diameter		0.65%	3.2	Gage
202	Package Height		0.65%	3.2	Gage
203	Evidence of Poor Workmanship		1.0%	3.16	Visual

NOTES:

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Screening Tests. - Screening tests required by 3.10.1 through 3.10.10 shall be performed for process control during the assembly of the hybrids. Performance of these tests shall be verified by in-process audit and review of data records required by 3.14. Bond Strength testing procedure shall be in accordance with MIL-STD-883, Method 5008, Paragraph 3.4.2, except the sampling interval shall not be greater than two (2) hours.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the examinations, and tests prescribed herein is described in the Paragraph Reference Inspection Method column in the tables of Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Electrical Test Procedures. - Electrical tests (4.5.1.1 through 4.5.1.5) shall be performed using the electrical test fixture (schematic diagram of Figure 1) in conjunction with the test equipment setup (Figure 2).

All alignment shall be performed at room ambient temperature using nominal supply and bias voltages of three and twenty five hundredths volts direct current plus or minus one hundredth volt direct current (3.25 VDC ± .01 VDC) and one hundred fifty five thousands volt direct current plus or minus one thousandth volt direct current (1.155 VDC ± .001 VDC) respectively. Adjustment or alignment to meet specification is not permitted for the temperature and voltage variation tests.

All test fixtures, procedures, and equipment shall be approved by the procuring activity.

4.5.1.1 Intermediate Frequency (IF) Gain.- The module shall be tested for the specified output signal at the IF output (J5) of the test fixture. A one millivolt (1 mv) maximum signal with a frequency of two million nine hundred thousand hertz plus or minus one thousand hertz (2,900,000 Hz ± 1000 Hz) shall be supplied to the IF input (J1) of the test fixture. Any module failing to comply with the specified IF gain requirement shall be classed defective (non-destructive test).

4.5.1.2 Discriminator Filter Characteristics.- The module shall be measured for the specified requirements at the discriminator filter output test point, pin 3 (J8). The module shall be supplied with a twenty five hertz (25 hz) sine wave at input test point (J6), and the amplitude

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at pin 3 (J8) noted as a reference. The voltage level shall be approximately thirty millivolts root mean square (30mv RMS). Increase the frequency of the sine wave generator until the amplitude at pin 3 (J8) is minus three decibels (-3 db) of the twenty five Hertz (25 Hz) response and note the frequency at which this occurs. (The amplitude of the signal at input J6 shall be kept constant). Any module failing to meet the response requirement shall be classified defective (non-destructive test).

4.5.1.3 Squaring Amplifier Gain. - The module shall be tested by measuring the signal gain between pin 3 (J8 of the test fixture) and pin 2 (J7 of the test fixture). A sine wave signal of two hundred and thirty Hertz plus or minus two Hertz (230 Hz ± 2 Hz) shall be applied to the discriminator filter input (J6) and its amplitude adjusted such that a signal of two and five tenths millivolts (2.5 mv RMS) maximum appears at the squaring amplifier input (J8 of the test fixture). The output is measured at pin 2 (J7). Any module failing to comply with the specified requirement shall be classified as defective (non-destructive test).

4.5.1.4 Squaring Amplifier Characteristics. - The module shall be tested utilizing the test setup of 4.5.1.3. A sine wave signal of two hundred and thirty Hertz plus or minus two Hertz (230 Hz ± 2 Hz) shall be applied to the discriminator filter input (J6) and its amplitude adjusted such that a sine wave signal with an amplitude of thirty five plus or minus three millivolts (35 ± 3 mv RMS) appears at squaring amplifier input (J8). The squaring amplifier characteristics shall be measured at the squaring amplifier output J7. Any module failing to comply with the specified requirement shall be classed as defective (non-destructive test).

4.5.1.5 Current Drain. - The Module shall be tested for the specified requirements utilizing the test fixture in the test setup with no signal applied to the test fixture input. Any module failing to comply with specified requirements shall be classified as defective (non-destructive test).

4.5.1.6 Input Capacitance. - The module shall be measured for the specified requirement with a capacitive meter having an accuracy of .25% or better. The input capacity measurement can be performed by placing the appropriate pins of the module directly to the measurement terminals of capacitance meter.

Pins Thirteen (13) and five (5) shall be connected together when measuring the capacitance between pins thirteen (13) and fifteen (15). Pin fifteen (15) shall be connected to the plus terminal and pin thirteen (13) to the minus terminal of the measuring equipment. The test signal used for measuring shall be kept at a low enough level to avoid forward biasing of the transistor. The capacity of a test fixture utilized to facilitate this measurement shall be subtracted from the measurement or compensated by an accepted technique. Any module failing to comply with the specified requirements shall be classified as defective (non-destructive test).

4.5.1.7 Voltage Variations. - The module shall be tested for the specified requirements in the test circuit with the supply and bias

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voltages varied to the specified limits of plus three and twenty five hundredths volts direct current plus and minus twenty five hundredths volt direct current (+3.25 VDC ± .25 VDC) and with the bias voltages and the corresponding tests shown in Table II.

TABLE II
VOLTAGE VARIATION TEST

Supply Voltage	Bias Voltage Test	
	1.128	1.158
3.00	4.5.1.1 4.5.1.3 4.5.1.5	4.5.1.1 4.5.1.3
	1.165	1.195
3.50	4.5.1.1 4.5.1.3	4.5.1.1 4.5.1.3

Any module failing to comply with all of the specified requirements shall be classified as defective (non-destructive test).

4.5.2 Temperature Variations. - The module shall be tested in the test circuit (figure 1) for specified requirements over the operating temperature range. Tests shall be conducted at temperatures of minus thirty two degrees centigrade (-32°C) maximum, zero degrees centigrade plus or minus five degrees centigrade (0°C ± 5°C), and plus sixty five degrees centigrade (+65°C) minimum. The bias voltage supplied to the module is a function of temperature and the supply voltage. The bias voltage, the supply voltage, the temperature, and the tests to be performed are shown in Table III.

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TABLE III
TEMPERATURE AND VOLTAGE VARIATION TEST

TEMP	SUPPLY VOLTAGE	LOW BIAS VOLTAGE VDC	HIGH BIAS VOLTAGE VDC
		TEST REQ.	TEST REQ.
-32°C Maximum	+ 3.00 VDC	1.160	1.190
		4.5.1.1 4.5.1.3	4.5.1.1 4.5.1.3 4.5.1.5
	+ 3.50 VDC	1.200	1.230
		4.5.1.1 4.5.1.3	4.5.1.1 4.5.1.3 4.5.1.5
0°C ± 5°C	+ 3.00 VDC	1.145	1.175
		4.5.1.1 4.5.1.3	4.5.1.1 4.5.1.3
	+ 3.50 VDC	1.180	1.210
		4.5.1.1 4.5.1.3	4.5.1.1 4.5.1.3 4.5.1.5
+65°C Minimum	+ 3.00 VDC	1.095	1.125
		4.5.1.1 4.5.1.3	4.5.1.1 4.5.1.3
	+ 3.50 VDC	1.135	1.165
		4.5.1.1 4.5.1.3	4.5.1.1 4.5.1.2 4.5.1.3 4.5.1.4 4.5.1.5

Any module failing to comply with all of the specified requirements shall be classified as defective (non-destructive test).

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4.5.3 Rework or Retest. - Any module failing to meet all of the requirements and is reworked or repaired shall be subject to all of the test performances of 4.5.1 before being classed acceptable. Any module failing to comply with all of the requirements after rework shall be classed defective. All rework or repair shall be documented as to cause of rejection and the work performed. The rejected data shall accompany the data for acceptance of the module (non-destructive test).

4.5.5 Data Requirements. - The contractor shall submit a quality inspection test, demonstration and evaluation report in accordance with Data Item Description DI-R-1724 (tailored) giving all detailed inspection and test results. Data shall include, but not be limited to, electrical performance test results.

5. PACKAGING

5.1 Preservation-packaging, packing and container marking shall be in accordance with level C requirements of MIL-M-55565 unless otherwise specified in the procurement document.

19

5.2 When modules of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6.0 NOTES

6.1 Intended Use. - The components and assemblies covered by this specification are intended for use as the IF amplifier for the XM122 firing device receiver. This specification is to be used only as a guideline in establishing on updated version of this device.

6.2 Ordering data. - See Section 6.1 and 6.2 of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval (See MIL-A-48078). - Submit equipment designs as required to: Commander, U.S. Army Armaments Research and Development Command, Dover, New Jersey, 07801, ATTN: DRDAR-QAA-I.

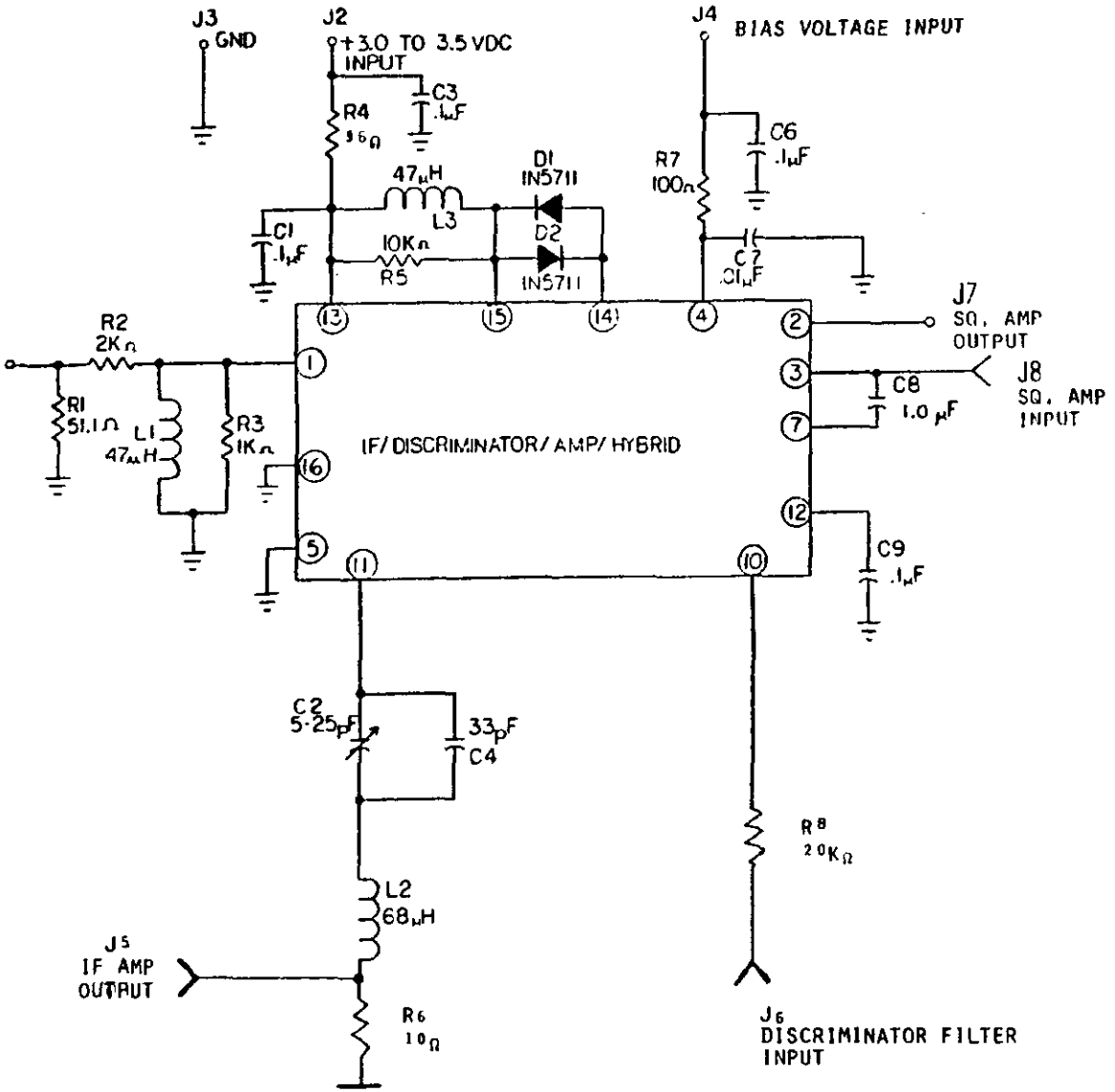
6.4 Distribution of ammunition data cards, inspection results and test data. - A copy of all ammunition data cards, inspection and test results shall be submitted to: Commander, U.S. Army Armament Research and Development Command, Dover, New Jersey 07801, ATTN: DRDAR-QAA-Q.

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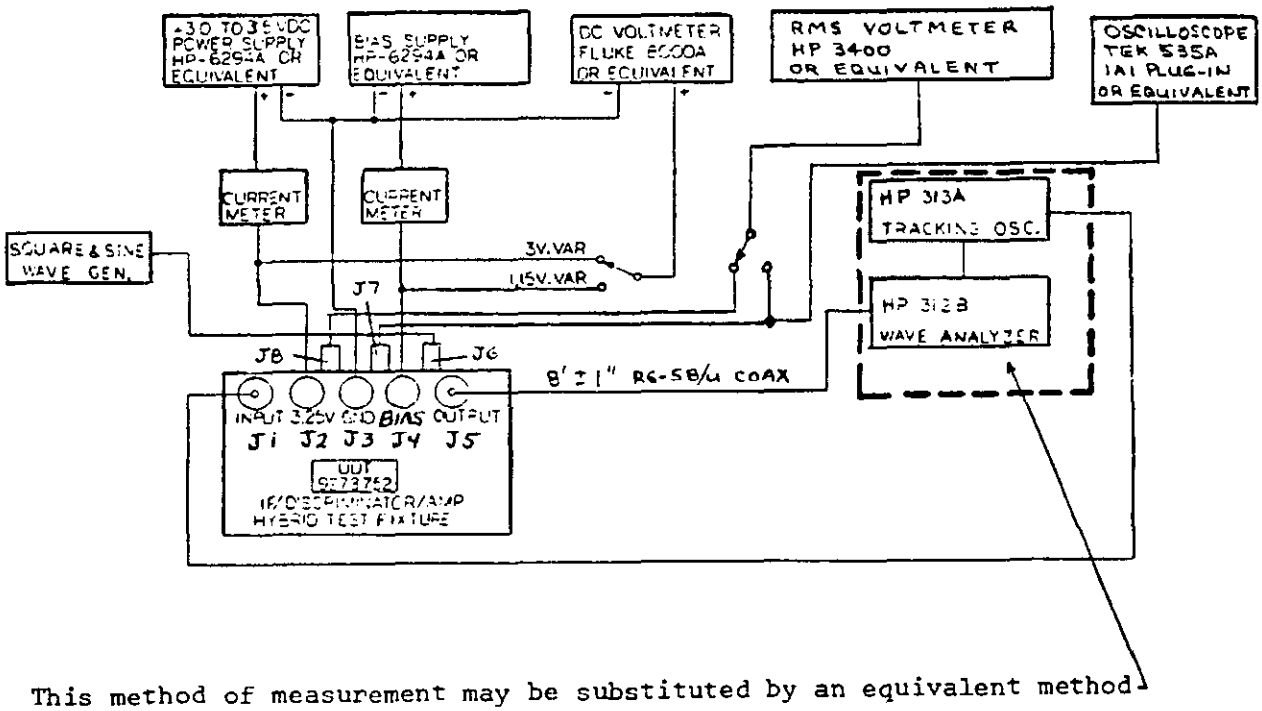
PARTS LIST/TEST FIXTURE

PART		REFERENCE DESIGNATION
CMOAED330JP3		C4
CK05BX103K		C7
CK06BX105K		C8
CK05BX104K		C1, 3, 6, 9
CV35A250		C2
RN55D51R1F		R1
RN55D1000F		R7
RN55D1001F		R3
RN55D2001F		R2
RN55D1002F		R6
RN55D1002F		R5
MS75085-03		L1, 3
MS21424-11		L2
IN5711		D1, 2,
9278786		Y1
111-0102-001	RED	J2
111-0103-001	BLACK	J3
111-0104-001	GREEN	J4
UG1094U		J1, 5, 6, 7, 8
RN55D2002F		R8
RN55D35R7F		R4

Figure 1. Test Fixture I[illegible]isc./Sg. Amp.

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TEST EQUIPMENT SETUP



This method of measurement may be substituted by an equivalent method

Figure 2. Test Equipment Setup-IF/Disc/Sq. Amp.

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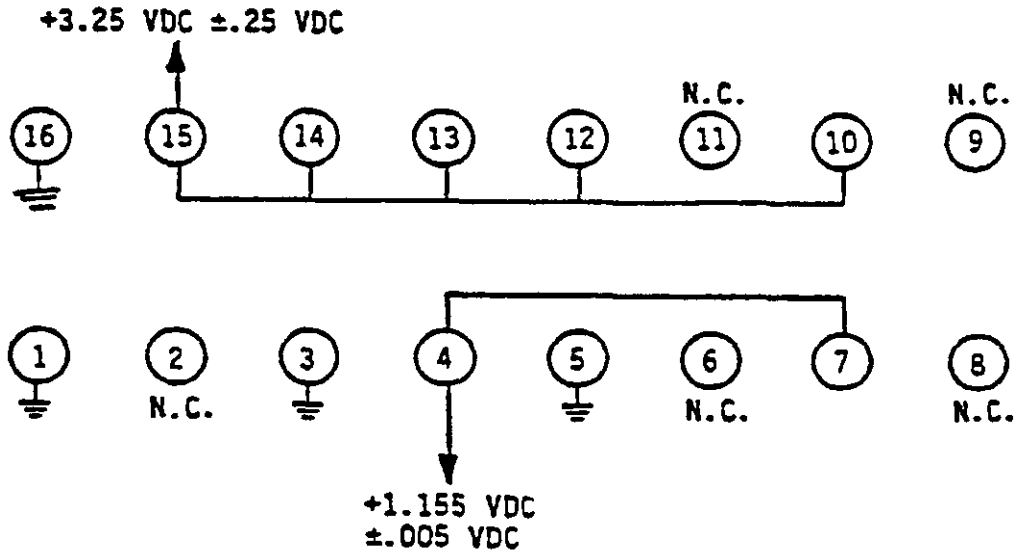


Figure 3. Burn-in Circuit for IF/Disc/Sq. Amp.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-63120A Intermediate Frequency/Discriminator/ Square Amplifier Hybr

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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